

EDUCATION

Ph.D. Mechanical Engineering and Applied Mechanics University of Pennsylvania	Anticipated May 2026 Cumulative GPA: 3.59
M.S. Mechanical Engineering and Applied Mechanics University of Pennsylvania	Graduated May 2024 Cumulative GPA: 3.6
B.S. Mechanical Engineering University of Maryland – College Park	Graduated May 2021 Cumulative GPA: 3.91

RESEARCH EXPERIENCE

PhD Student, Graduate Research Assistant

Architected Materials Laboratory Philadelphia, PA

August 2021 – Present

Advised by Prof. Jordan Raney | Trained in novel manufacturing methods to produce materials and structures with unparalleled engineering capabilities.

Project 1: Valveless pumping methods for restoring cavopulmonary Fontan conduit pulsatility.

- Uncovered key mechanisms governing valveless pumping, developed a numerical model to capture relevant nonlinear fluid dynamics, and performed guided experiments to study resonance-driven flow amplification.
- Innovated peristaltic conduit device designs with medical input from collaborating physicians from CHOP, using modeling and FEA tools, and theoretical concepts to optimize fluid dynamics and energy efficiency, while ensuring mechanical integrity and biocompatibility.
- Designed and fabricated an in vitro mock cavopulmonary circulation test setup for peristaltic device validation.

Project 2: Leveraging metamaterials for passive unsteady flow control.

- Explored interaction between mechanical metamaterials and turbulent flows to enhance energy efficiency and flight performance in DoD air vehicles, improving sustainability and operational capability.
- Ideated and fabricated passive vortex generators and a decambering airfoil mechanism that deploy upon sensing of critical flow signatures of separation.

Project 3: Experimental characterization and analytical modeling for granular jamming structures.

- Developed a model using elastic-plastic beam bending theory to characterize mechanical response beyond elastic regime and validated by incorporating experimental results from three-point bending tests.
- Defended independent research study characterizing granular jamming in soft actuators for therapeutic devices by delivering a comprehensive research paper, an oral presentation, and chalk talk.
- Provided a design guideline framework for development of adaptive medical devices and soft robotic systems requiring tunable rigidity and reconfigurable functionality.

Undergraduate Research Assistant

Tubaldi Lab College Park, MD

June 2020 – May 2021

Advised by Prof. Eleanora Tubaldi | Trained in theory, simulations, and experiments to explore systems at the intersection of nonlinear dynamics, soft materials, and fluid-structure interactions.

- Implemented cardiovascular arterial wall structure models to study their biomechanical properties and investigate fluid-structure interactions with Abaqus numerical analysis.
- Revised existing models and scripts and developed a manufacturing and testing plan for vascular structures.
- Fabricated soft material structures using silicone molding techniques and performed mechanical testing to validate theoretical predictions.

TECHNICAL EXPERIENCE

Mechanical Engineering Intern

Stanley Black & Decker Towson, MD

June 2021 – August 2021

- Identified pertinent test parameters, conducted in-lab testing on custom test jig, and analyzed data aiming to increase tool productivity by spending less time cutting and grinding with the angle grinder.
- Conducted a design of experiments (DOE) to investigate flap disc design parameters that may affect performance.

Pipelines and Risers Engineering Intern**ExxonMobil Houston, TX****August 2020 – December 2020**

- Developed and evaluated desktop engineering design tools using Excel and MATLAB to support the performance of in-house design calculation verification for various project phases.
- Communicated project findings virtually during team meetings with detailed milestone presentations following a self-developed weekly work plan.

Structural Engineering Intern**Boeing Everett, WA****May 2020 – August 2020**

- Collaborated with a team of interns, mentors, and managers to develop company rapid prototyping processes for additive manufacturing by designing (in CATIA) a load bearing fluid transport structure satisfying multiple conflicting design requirements, testing, and experimentally validating.
- Modified design through iteration after initial Abaqus FEM analysis to reduce stress in critical areas to prevent failure under loading.

Mechanical Engineering Intern**Textron Systems Hunt Valley, MD****June 2019 – August 2019**

- Oversaw program development tasks by supporting the testing and troubleshooting of processes and equipment.
- Designed and modeled a safety barrier to interface with the Common Unmanned Surface Vessel using NX 10.0 to support a list of requirements and performed analysis and calculations to assess capability of design.
- Drafted engineering drawings for parts and assemblies for use by Engineering Prototype Lab to manufacture non-deliverable and deliverable products currently being used in the lab.

TEACHING AND LEADERSHIP EXPERIENCE

Graduate Teaching Assistant**University of Pennsylvania Philadelphia, PA**

- Statics and Strength of Materials
- Thermodynamics I
- Statics and Strength of Materials

August 2023 – January 2024**January 2023 – May 2023****August 2022 – December 2022*****Undergraduate Teaching Assistant*****University of Maryland College Park, MD**

- Machine Design
- Dynamics

January 2021 – May 2021**January 2020 – May 2020*****President*****Mechanical Engineering Graduate Organization Philadelphia, PA****October 2022 – November 2023**

- Advocated for mechanical engineering graduate student body, facilitating change for qualifying exam procedures.
- Organized 13 social events engaging over 100 members to foster community and engagement among graduate students, faculty, and staff.
- Coordinated professional development seminars and mock qualifying exam sessions to assist junior cohort prepare for their comprehensive qualifying exam.

Aeroshell Team Member**UMD Loop College Park, MD****March 2019 – December 2020**

- Researched carbon fiber composite layup methods and sourced equipment to develop manufacturing plan.
- Conducted feasibility study, with a team of six, on designing, manufacturing, and testing a tunnel boring machine and its subsystems given limited project guidelines.

Project Lead, Team Member**Engineers Without Borders College Park, MD****December 2017 – May 2021**

- Exercised leadership skills to communicate with students, faculty advisors, mentors, and EWB USA to progress design implementation and project process timeline.
- Collaborated with sub-teams to design the implementation of photovoltaic systems in Sierra Leone and Ghana and traveled with team of students and mentors to implement designed systems.

TECHNICAL SKILLS

Wet Lab: Prepare customizable printing “inks” for direct ink writing, measuring and mechanical testing equipment (Instron, high speed cameras, microscopes, digital image correlation), additive manufacturing equipment (FDM, FFF, SLA, DIW), casting and molding

Modeling and Analysis: Siemens NX, Abaqus, Ansys APDL, Ansys Workbench, Autodesk Inventor, CATIA V5, Creo Parametric, SolidWorks

Programming: Arduino, C++, MATLAB, Python, R

Miscellaneous: Microsoft Office Suite, Teamcenter, mechatronics, circuit design, machine learning

Machining: Mill, lathe, band saw, belt sander, grinder, drill press

HONORS AND AWARDS

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| 1. Outstanding Service Award (University of Pennsylvania) | May 2024 |
| 2. GAPSA Individual Conference Travel Grant (University of Pennsylvania) | February 2023 |
| 3. Lisa P. Yu Fellowship (University of Pennsylvania) | August 2021 – August 2022 |
| 4. Graduated Cum Laude with Honors (University of Maryland) | May 2021 |
| 5. Lockheed Martin STEM Scholar (University of Maryland) | May 2019 – May 2021 |
| 6. Board of Visitors Endowed Scholarship (University of Maryland) | August 2019 – May 2021 |

PUBLICATIONS

1. He Q, Yin R, Hua Y, Shu H, Zhu X, Haque AT, Ferracin S, Patel S, Jiao W, Raney JR. “Modular Stimuli-Responsive Valves for Pneumatic Soft Robots”. *Advanced Intelligent Systems*. 2024:2400659.
2. *(Submitted)* Unger G, Patel S, Kozyak B, Raney J, Sung C, “Liebau Pumping Enables Valveless Swimmer Robot”. *International Symposium of Experimental Robotics*.
3. *(In preparation)* Patel S, Unger G, Kozyak B, Sung C, Raney J, “Valveless pumping as a mechanism for Fontan circulation augmentation”.

CONFERENCES

1. *(Expected abstract)* 2025 Society of Engineering Science (SES) Annual Technical Meeting: Controlling Mechanical Waves with Metamaterials
2. Gordon Research Conference: Optimizing the Synergy Between Architecture, Non-Equilibrium Processes and Materials 2023 *(Attendee)*
3. IEEE International Conference on Robotics and Automation 2022 *(Volunteer and attendee)*